Power Query Data Cleaning Guide

Chat Institute Student Dataset | 10,000 rows | Power BI Desktop

Step 1: Load the CSV into Power Query

1. Open Power BI Desktop.
2. Go to **Home** → **Get Data** → **Text/CSV**.
3. Select **chat_institute_students.csv**.
4. In the preview window, click **Transform Data** (not Load). This opens Power Query Editor.

Step 2: Turn on data profiling

1. Go to the **View** tab.
2. Tick **Column quality**, **Column distribution** and **Column profile**.
3. At the bottom-left status bar, click **"Column profiling based on top 1000 rows"** and change it to **"based on entire data set"**.

Each column now shows **Valid / Error / Empty** percentages, so you can spot blanks and bad values.

Step 3: Delete the auto "Changed Type" step

1. In **Applied Steps** (right pane), find **Changed Type**.
2. Click the **X** next to it to delete it.

Power BI guesses types automatically and gets some wrong. For example, **Phone** becomes a number. You will set every type yourself in Step 5.

Step 4: Trim and clean the text

1. Click any column header, then press **Ctrl + A** to select all columns.
2. Right-click a header → **Transform** → **Trim** (removes extra spaces).
3. Right-click again → **Transform** → **Clean** (removes hidden characters).

Step 5: Set the data types

Select columns with **Ctrl + click**, right-click → **Change Type**, or click the **ABC/123** icon on each header. If a popup asks **Replace current** or **Add new step**, choose **Replace current**.

Type	Columns
Text	Student_ID, Student_Name, Gender, Email, **Phone**, City, State, State_Code, College_Name, College_Code, College_City, College_State, **USN**, Degree, Department, Dept_Code, Student_Type, Institute_Name, Branch_Code, Branch_Name, Branch_City, Branch_State, Course_Code, Course_Name, Course_Category, Batch_ID, Batch_Mode, Batch_Timing, Trainer_Name, Enrollment_Month, Enrollment_Quarter, Payment_Mode, Payment_Status, Final_Grade, Course_Status, Certificate_Issued, Placement_Status, Company_Placed, Lead_Source
Date	Date_of_Birth, Enrollment_Date, Course_Start_Date, Course_End_Date, Placement_Date
Whole Number	Age, Year_of_Passout, Course_Duration_Months, Enrollment_Year, Discount_Percent, Installments, Feedback_Rating
Fixed Decimal (Currency)	Course_Fee, Discount_Amount, Final_Fee, Fees_Paid, Fees_Due
Decimal Number	CGPA, Attendance_Percent, Assessment_Score, Project_Score, Package_LPA

Rule of thumb: if you will never do math on it (phone, USN, IDs, codes), make it **Text**.

Step 6: Check for errors

1. Look at the **Column quality** bar on date and number columns. **Error** should be 0%.
2. If any column shows errors, right-click its header → **Replace Errors** → type **null**.

Step 7: Handle blank values

Text blanks: replace them with meaningful labels.

1. Right-click **Company_Placed** → **Replace Values**. Leave "Value To Find" **empty**, set "Replace With" to **Not Applicable**.
2. Right-click **Final_Grade** → **Replace Values**. Leave "Value To Find" empty, set "Replace With" to **Not Graded**.

Number and date blanks: leave them as **null**. Do **not** replace with 0. Columns: Attendance_Percent, Assessment_Score, Project_Score, Package_LPA, Feedback_Rating, Placement_Date.

Why this matters: an ongoing student has no score yet. If you put 0, the average assessment score drops sharply and every KPI becomes wrong. AVERAGE skips nulls but counts zeros.

Step 8: Standardize text formatting

1. Right-click **Student_Name** → **Transform** → **Capitalize Each Word**.
2. Right-click **Email** → **Transform** → **lowercase**.

Step 9: Remove duplicates

1. Right-click **Student_ID** → **Remove Duplicates**.
2. Check the row count at the bottom-left. It should still be **10,000** (this step is a safety check).

Step 10: Add helper columns

Go to **Add Column** → **Custom Column** for each one. After creating it, set the type shown.

New Column	Formula	Type
Fee_Collection_Pct	`Number.Round([Fees_Paid] / [Final_Fee] * 100, 2)`	Decimal
Age_Group	`if [Age] <= 22 then "18-22" else if [Age] <= 25 then "23-25" else "26+"`	Text
CGPA_Band	`if [CGPA] >= 9 then "9-10" else if [CGPA] >= 8 then "8-9" else if [CGPA] >= 7 then "7-8" else if [CGPA] >= 6 then "6-7" else "Below 6"`	Text
Is_Placed	`if [Placement_Status] = "Placed" then 1 else 0`	Whole Number
Days_To_Placement	`if [Placement_Date] = null then null else Duration.Days([Placement_Date] - [Course_End_Date])`	Whole Number
Enrollment_Month_No	`Date.Month([Enrollment_Date])`	Whole Number

Month sorting: after loading, in Data view select **Enrollment_Month** → **Column tools** → **Sort by column** → **Enrollment_Month_No**. Otherwise months sort alphabetically (Apr, Aug, Dec...).

Step 11: Rename and load

1. In the left **Queries** pane, right-click the query → **Rename** → **Students**.

2. Click **Home** → **Close & Apply**.

Expected result: **10,000 rows**, **67 columns** (61 original + 6 new), **0 errors**.

Shortcut: All Steps as One M Script

Go to **Home** → **Advanced Editor**, delete everything, paste this script and change only the file path on line 2.

```
let
    Source = Csv.Document(File.Contents("C:\Users\YourName\Downloads\chat_institute_students.csv"),
        [Delimiter=",", Columns=61, Encoding=65001, QuoteStyle=QuoteStyle.Csv]),
    PromotedHeaders = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),

    TrimmedCleaned = Table.TransformColumns(
        PromotedHeaders,
        List.Transform(Table.ColumnNames(PromotedHeaders),
            each {_, each Text.Clean(Text.Trim(_)), type text})
    ),

    ChangedTypes = Table.TransformColumnTypes(TrimmedCleaned, {
        {"Date_of_Birth", type date}, {"Enrollment_Date", type date}, {"Course_Start_Date", type date},
        {"Course_End_Date", type date}, {"Placement_Date", type date},
        {"Age", Int64.Type}, {"Year_of_Passout", Int64.Type}, {"Course_Duration_Months", Int64.Type},
        {"Enrollment_Year", Int64.Type}, {"Discount_Percent", Int64.Type}, {"Installments", Int64.Type},
        {"Feedback_Rating", Int64.Type},
        {"Course_Fee", Currency.Type}, {"Discount_Amount", Currency.Type}, {"Final_Fee", Currency.Type},
        {"Fees_Paid", Currency.Type}, {"Fees_Due", Currency.Type},
        {"CGPA", type number}, {"Attendance_Percent", type number}, {"Assessment_Score", type number},
        {"Project_Score", type number}, {"Package_LPA", type number}
    }, "en-US"),

    ErrorsToNull = Table.ReplaceErrorValues(ChangedTypes, {
        {"Placement_Date", null}, {"Attendance_Percent", null}, {"Assessment_Score", null},
        {"Project_Score", null}, {"Package_LPA", null}, {"Feedback_Rating", null}
    }),

    FixCompany = Table.ReplaceValue(ErrorsToNull, "", "Not Applicable",
        Replacer.ReplaceValue, {"Company_Placed"}),
    FixGrade = Table.ReplaceValue(FixCompany, "", "Not Graded",
        Replacer.ReplaceValue, {"Final_Grade"}),

    ProperNames = Table.TransformColumns(FixGrade, {
        {"Student_Name", Text.Proper, type text},
        {"Email", Text.Lower, type text}
    }),

    RemovedDuplicates = Table.Distinct(ProperNames, {"Student_ID"}),

    AddFeePct = Table.AddColumn(RemovedDuplicates, "Fee_Collection_Pct",
        each Number.Round([Fees_Paid] / [Final_Fee] * 100, 2), type number),
    AddAgeGroup = Table.AddColumn(AddFeePct, "Age_Group",
        each if [Age] <= 22 then "18-22" else if [Age] <= 25 then "23-25" else "26+", type text),
    AddCGPABand = Table.AddColumn(AddAgeGroup, "CGPA_Band",
        each if [CGPA] >= 9 then "9-10" else if [CGPA] >= 8 then "8-9" else if [CGPA] >= 7 then "7-8"
             else if [CGPA] >= 6 then "6-7" else "Below 6", type text),
    AddIsPlaced = Table.AddColumn(AddCGPABand, "Is_Placed",
        each if [Placement_Status] = "Placed" then 1 else 0, Int64.Type),
    AddDaysToPlace = Table.AddColumn(AddIsPlaced, "Days_To_Placement",
        each if [Placement_Date] = null then null
             else Duration.Days([Placement_Date] - [Course_End_Date]), Int64.Type),
    AddMonthNo = Table.AddColumn(AddDaysToPlace, "Enrollment_Month_No",
        each Date.Month([Enrollment_Date]), Int64.Type)
in
    AddMonthNo
```